

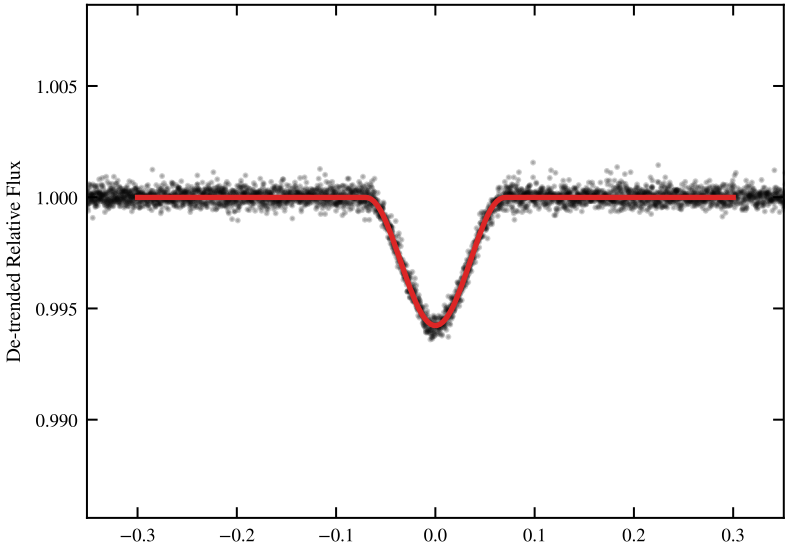
TESS PLANET TRANSIT CHARACTERIZATION REPORT

Target: TIC 128790976 | RA: 300.21293° | Dec: -43.98253° | Sectors: 13

Derived Planetary Parameters

Parameter	Derived Value (MCMC)
Period (d)	3.5193 ± 9.65e-05
Rp/R★	0.33722 ± 0.0943
Rp (R⊕)	55.065 ± 15.5
b	1.1687 ± 0.105
T14 (h)	3.3694 ± 0.053
a (AU)	0.037209 ± 0.00134
Teq (K)	2044.1 ± 36.5

Primary Phased Transit



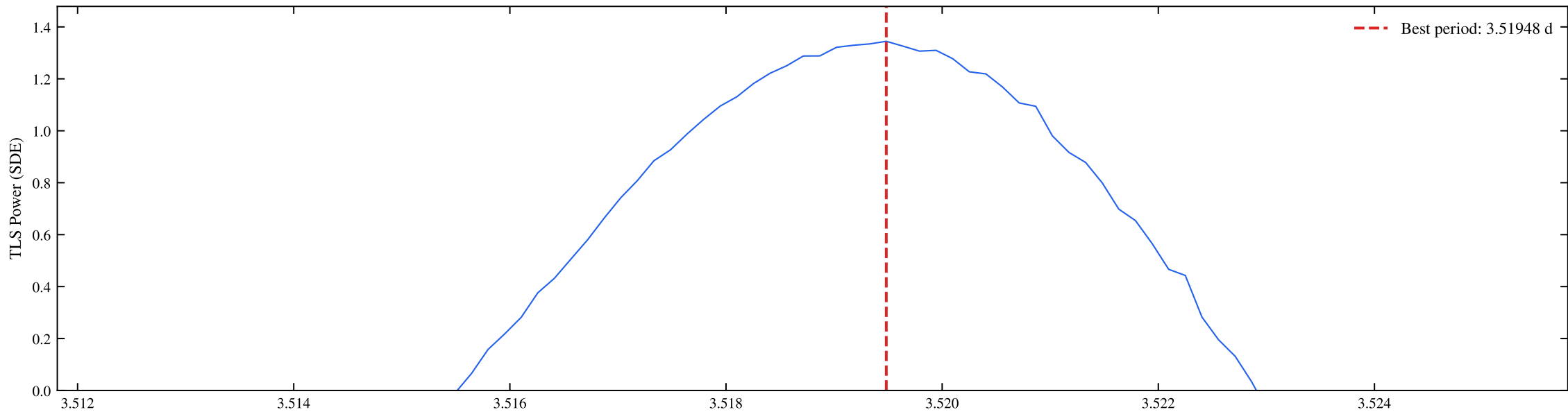
Literature Comparison & MCMC Diagnostics

Parameter / Metric	Literature Value	Derived / Diagnostic
P1 Period (d)	3.51948	3.519344 ± 0.00010
P1 Epoch (BTJD)	N/A	1657.4356 ± 0.0004
Max R-hat (R)	N/A	1.0068
Min ESS	N/A	602
MCMC Divergences	N/A	139

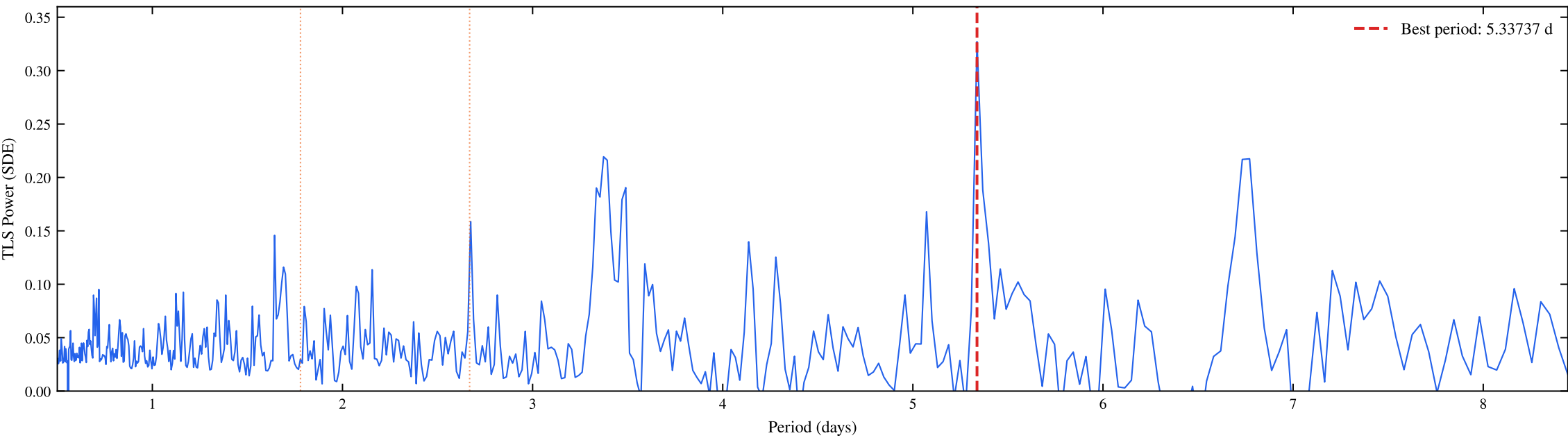
Host Star Properties

Stellar Parameter	Value	Source
R★ (R☉)	1.495 ± 0.048	VizieR TIC8
M★ (M☉)	1.42 ± 0.207	VizieR TIC8
Teff (K)	6690 ± 106	VizieR TIC8
log g (dex)	4.241 ± 0.0646	VizieR TIC8
[Fe/H]	-0.3832 ± 0.013	Gaia DR3
ρ★ (g/cm³)	0.5991 ± 0.105	Derived

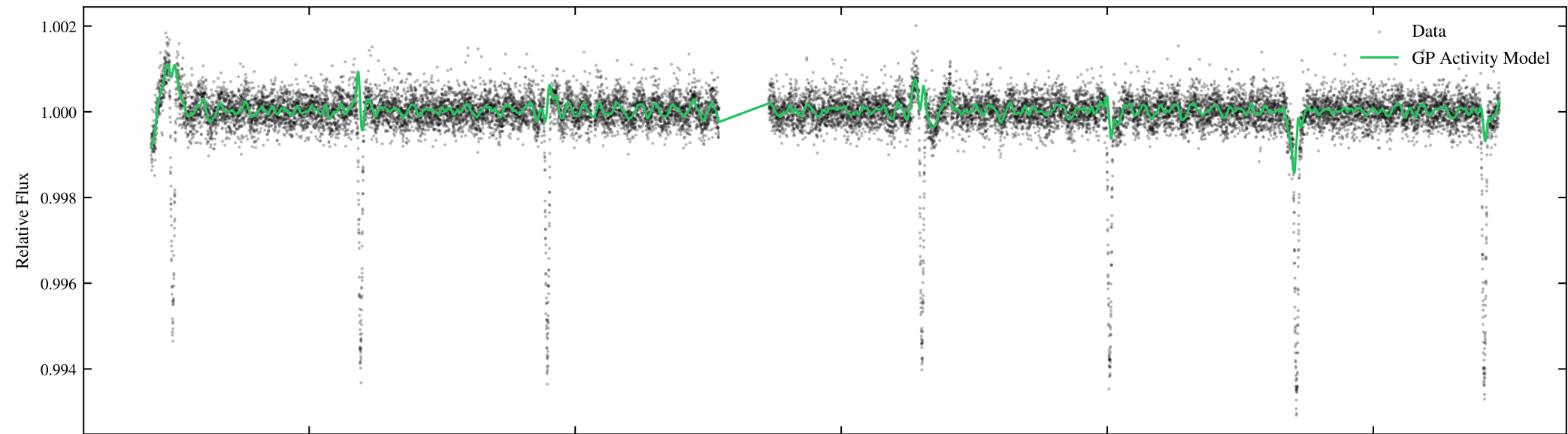
TIC 128790976 | Sectors: 13 | TLS Periodogram (Planet 1 Search) (SDE = 1.34)



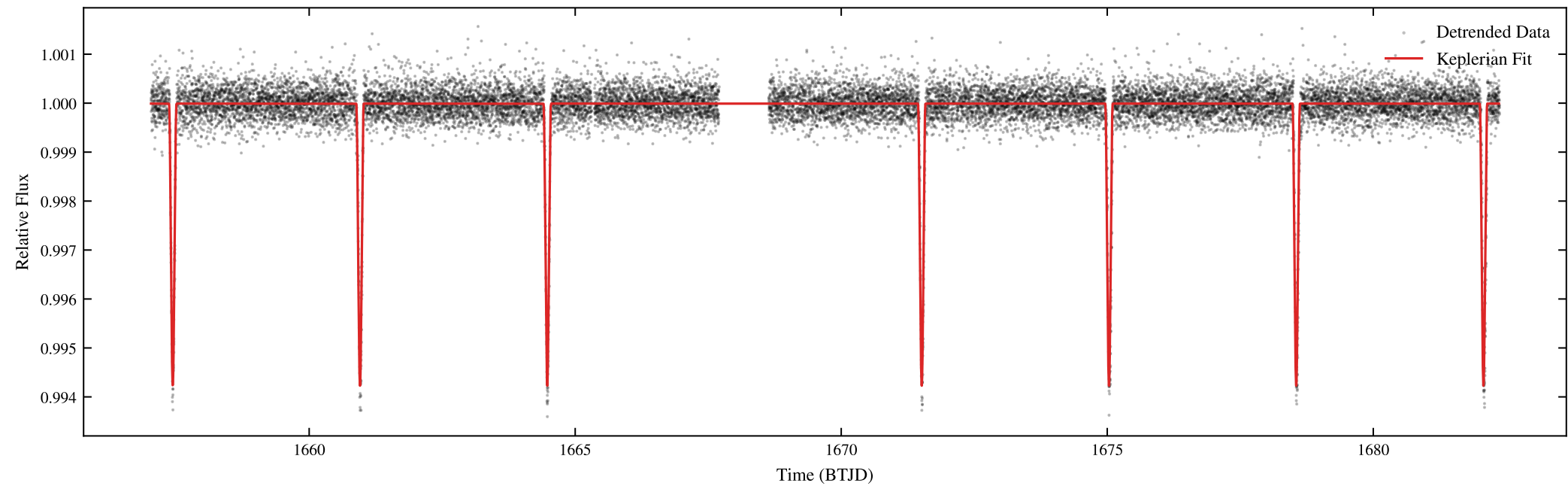
TIC 128790976 | Sectors: 13 | TLS Periodogram (Planet 2 Search) (SDE = 0.33)



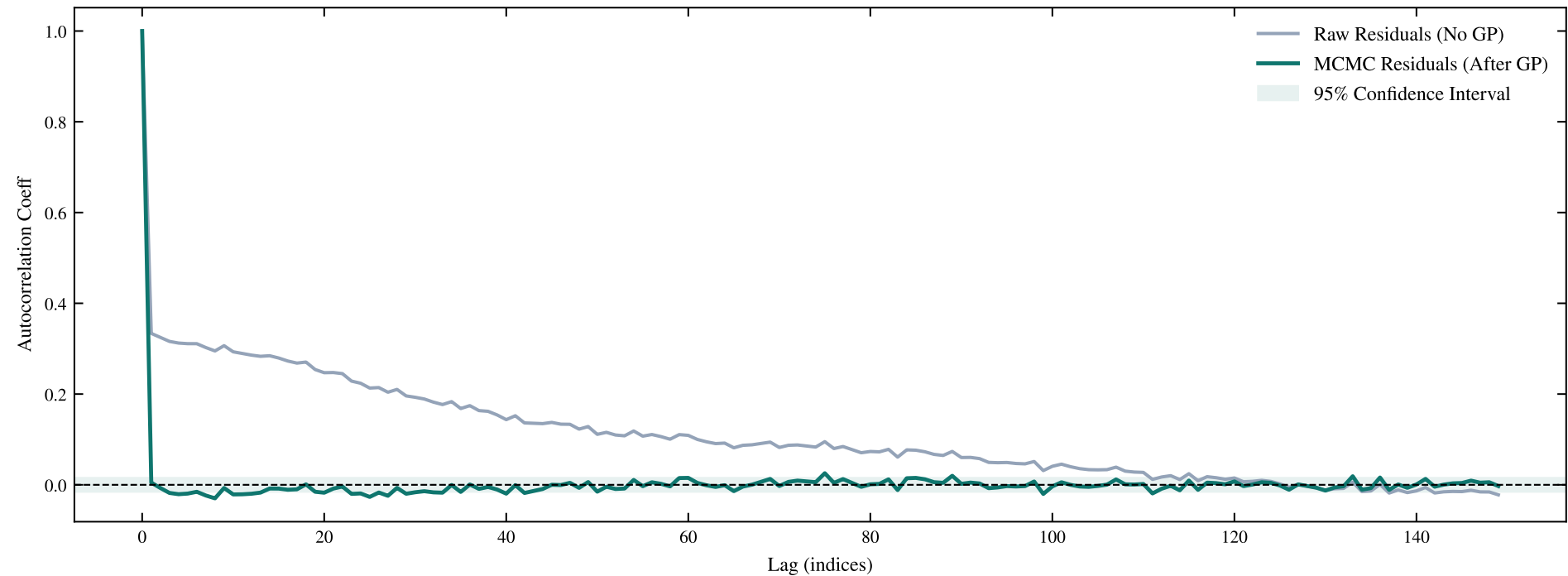
TIC 128790976 | Sectors: 13 | Bayesian GP Detrending (Time Domain)



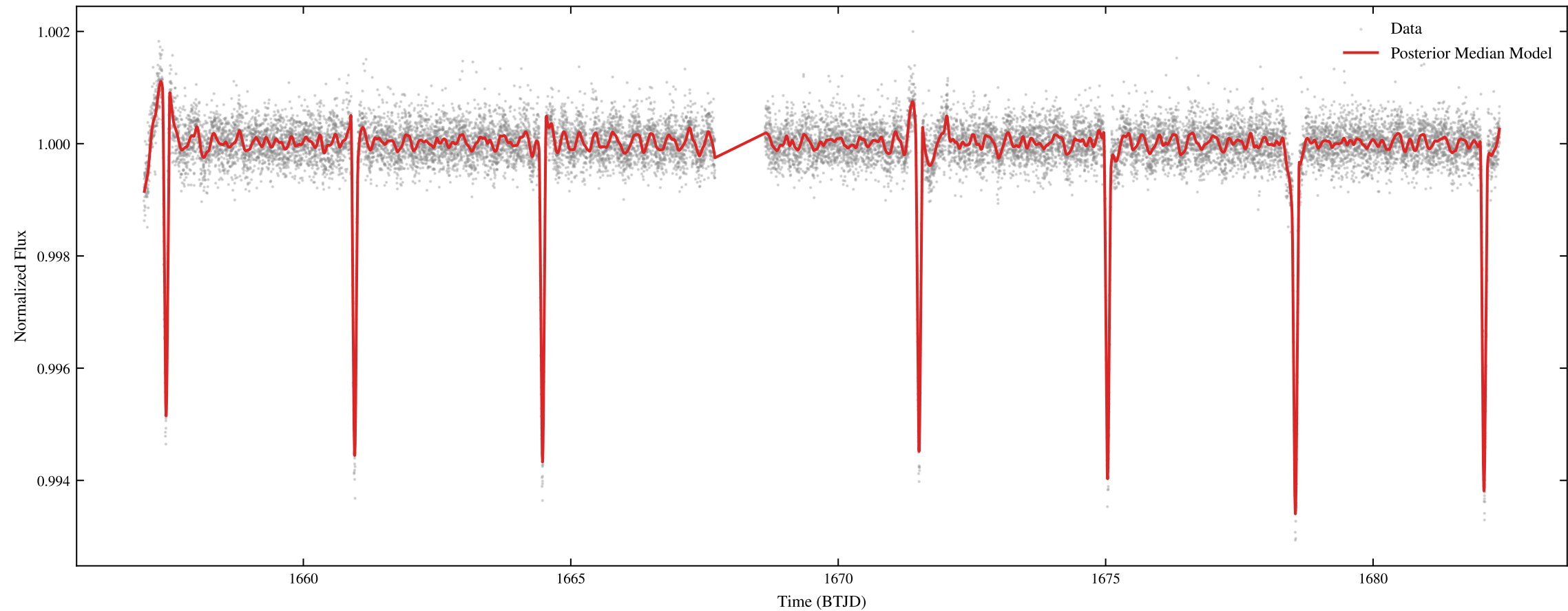
Detrended Light Curve & Best-Fit Transit Model

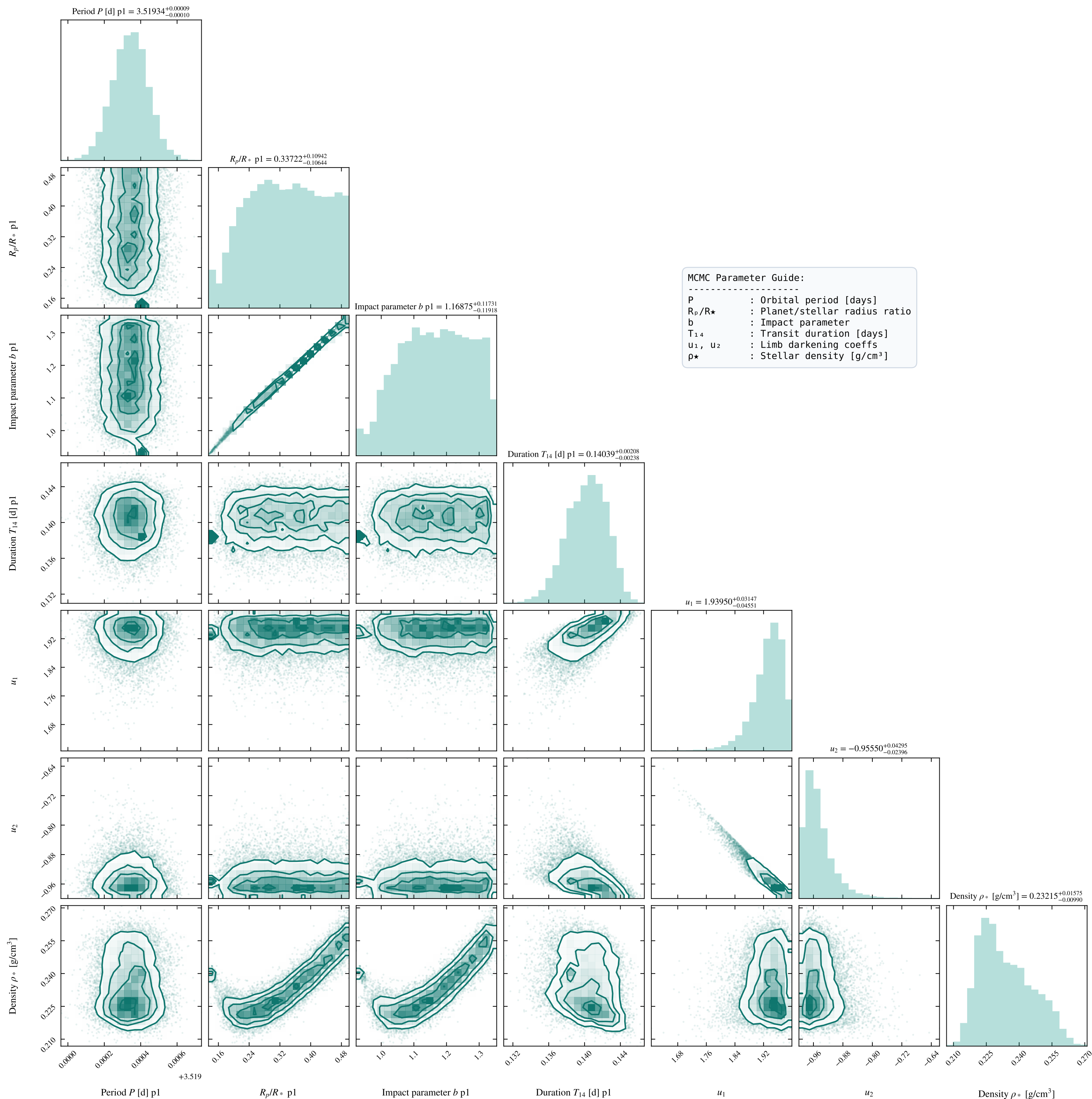


TIC 128790976 | Sectors: 13 | GP Residuals Autocorrelation Function (ACF)



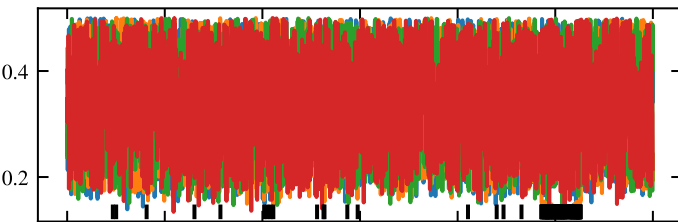
TIC 128790976 | Sectors: 13 | MCMC Posterior Predictive Check





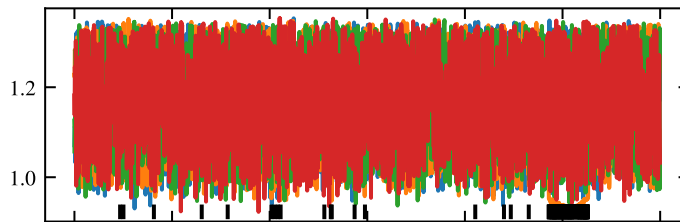
TIC 128790976 | Sectors: 13 | MCMC Trace Plots

rp_r_star
0



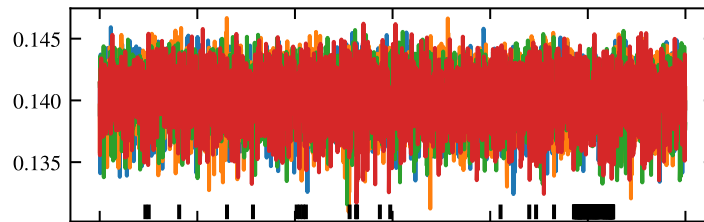
Draw

b
0



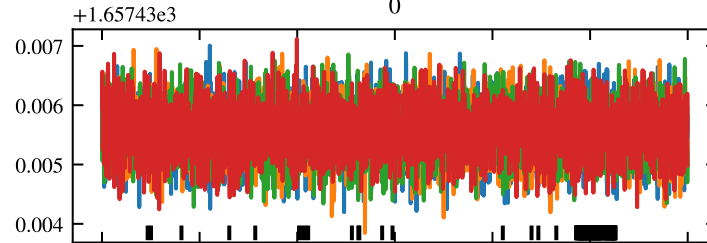
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t14
0



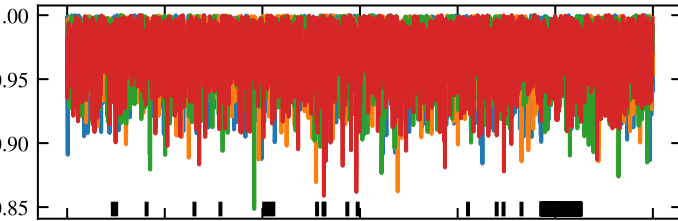
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t0
0



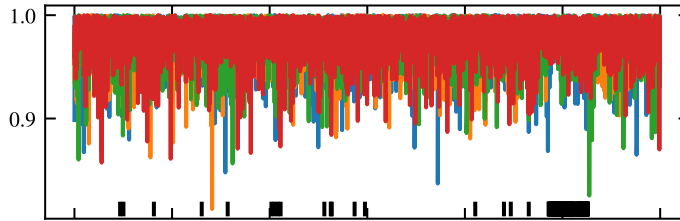
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q1



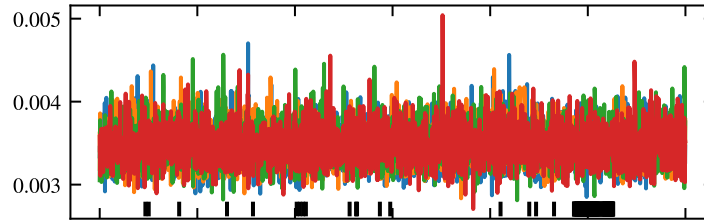
Draw

q2



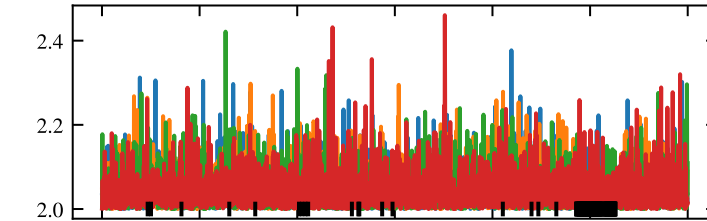
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sigma_gp



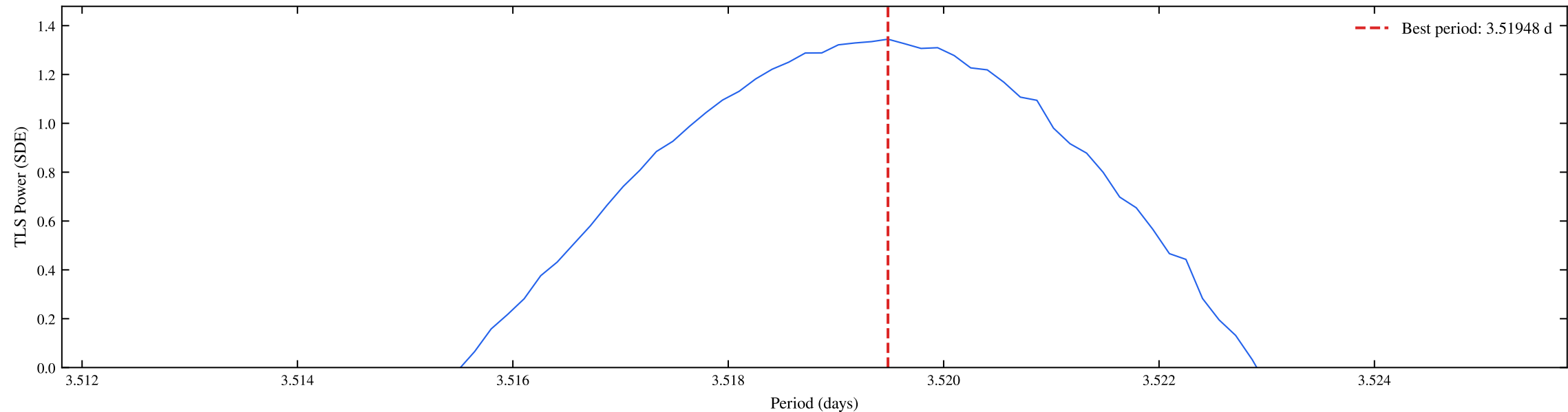
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rho_gp

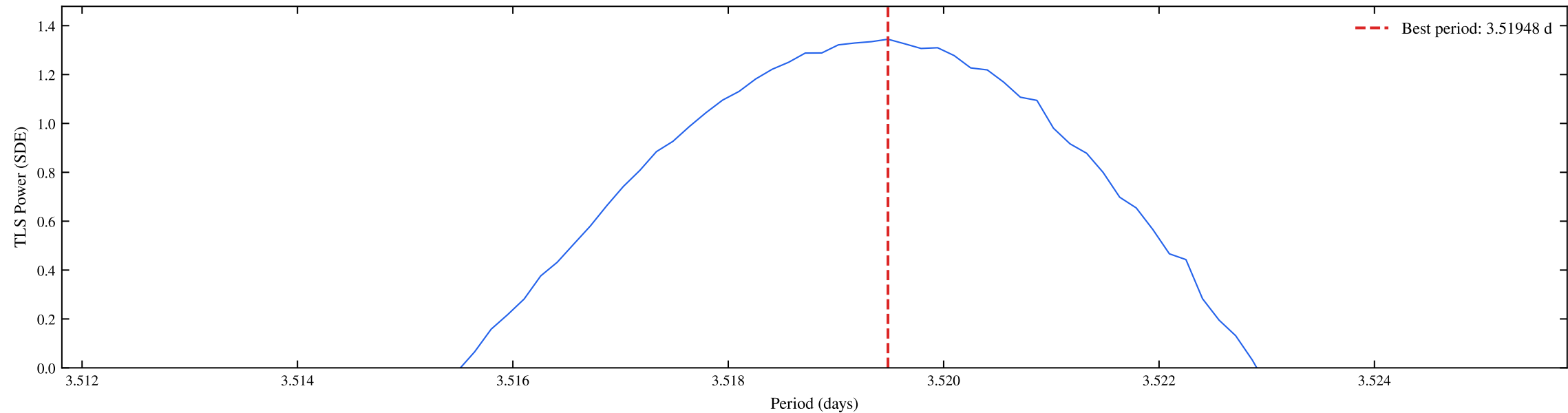


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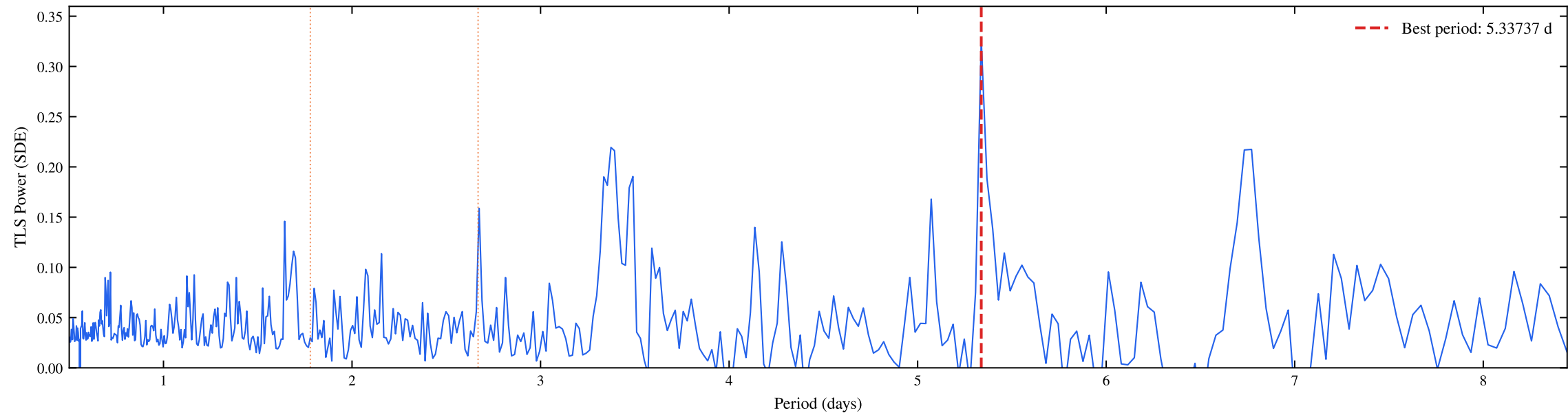
TIC 128790976 | Sectors: 13 | TLS Coarse Periodogram (SDE = 1.34)



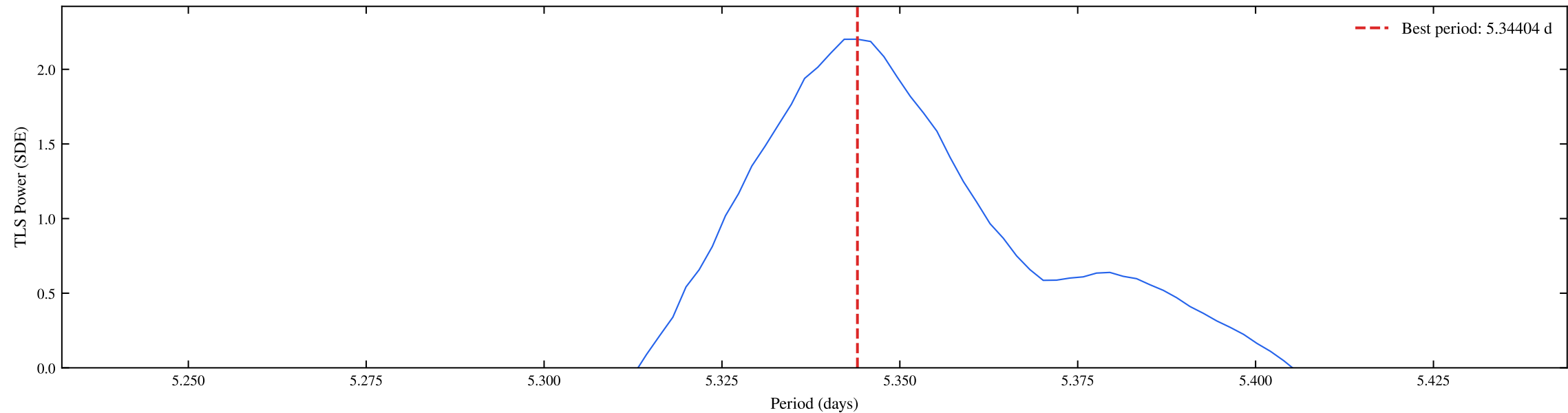
TIC 128790976 | Sectors: 13 | TLS Fine Periodogram (SDE = 1.34)



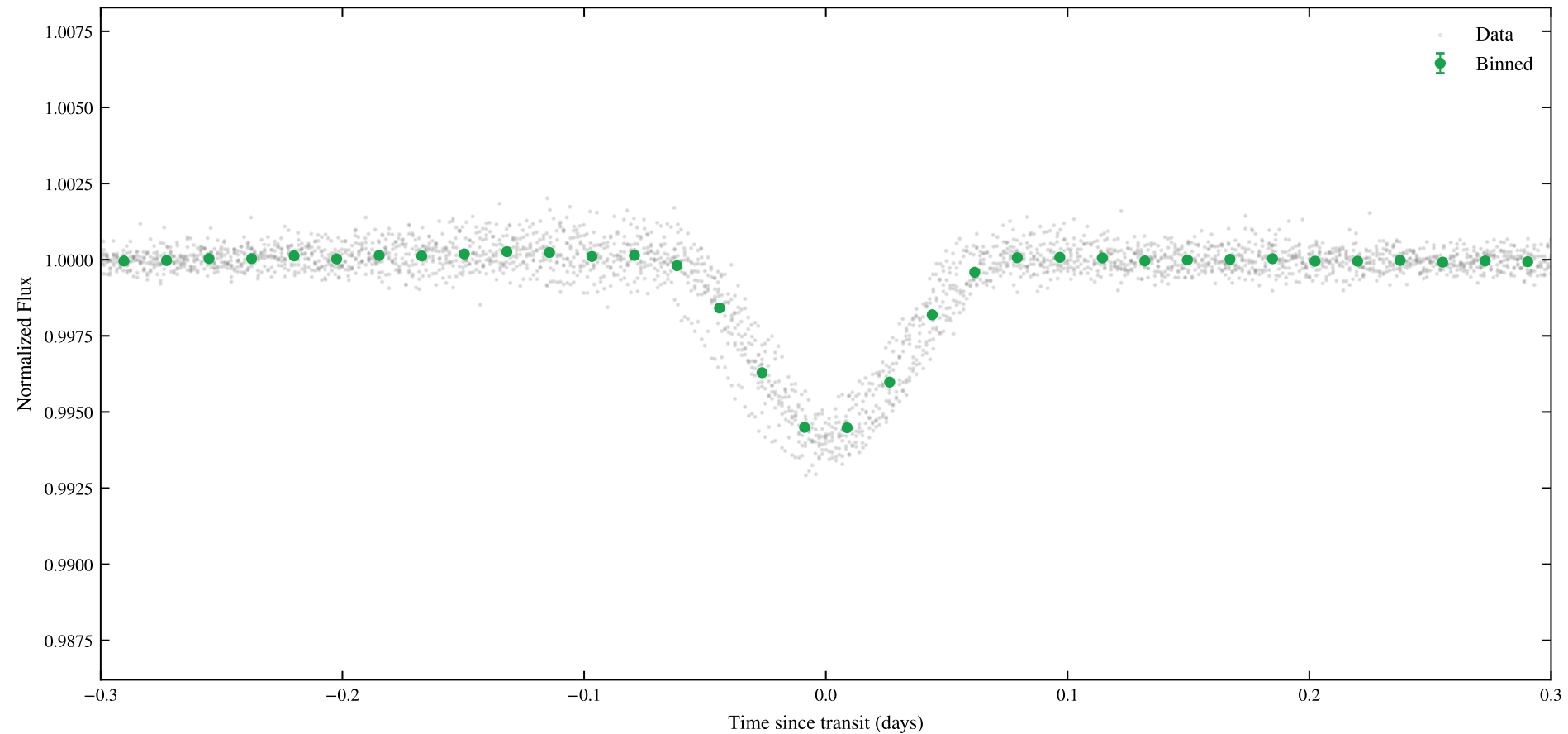
TIC 128790976 | Sectors: 13 | TLS Coarse Periodogram (SDE = 0.33)



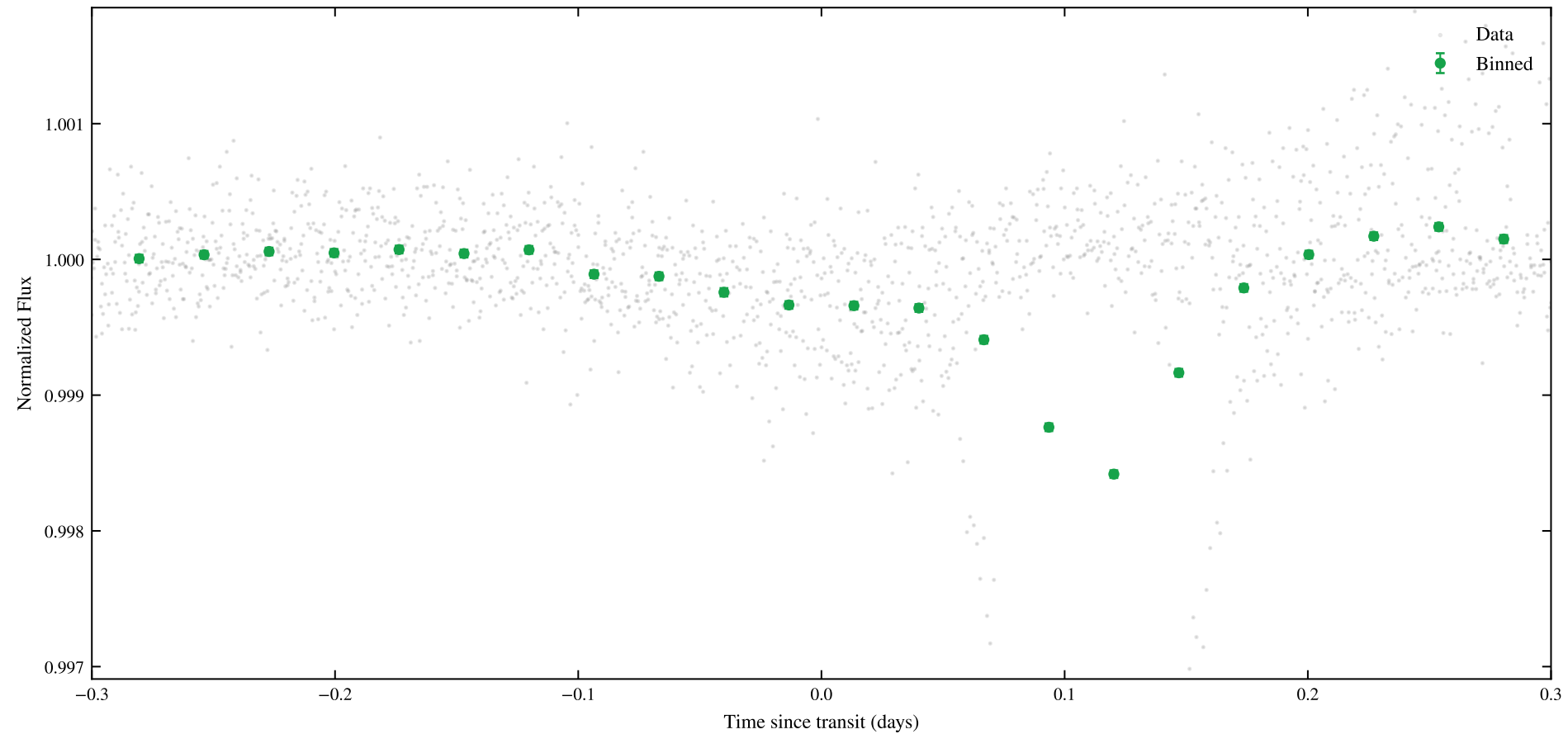
TIC 128790976 | Sectors: 13 | TLS Fine Periodogram (SDE = 2.20)



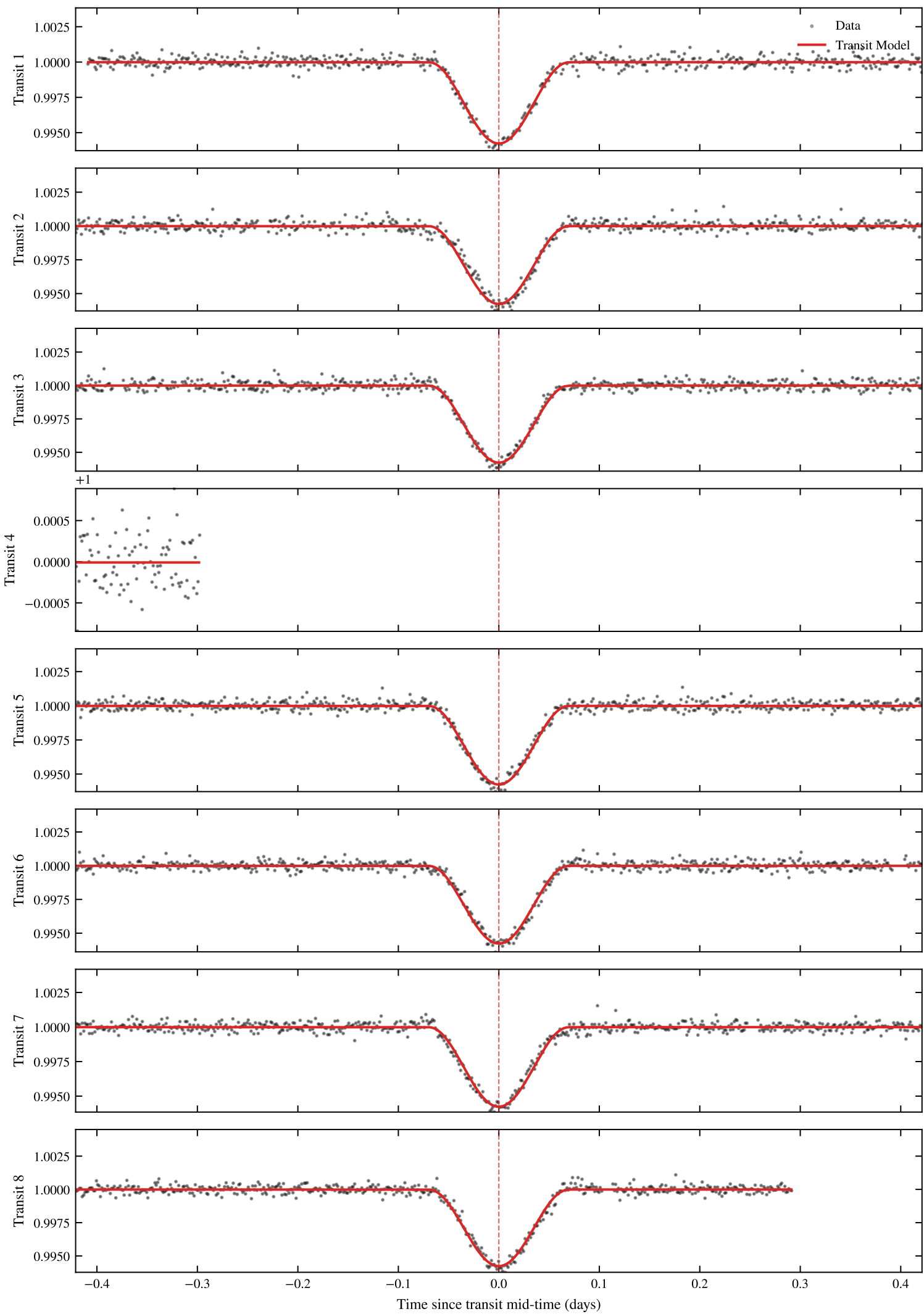
TIC 128790976 | Sectors: 13 | Phase-Folded Transit (P = 3.51948 d)



TIC 128790976 | Sectors: 13 | Phase-Folded Transit (P = 5.34404 d)



TIC 128790976 | Sectors: 13 | Stacked Individual Transits



TIC 128790976 | Sectors: 13 | Phased MCMC Fit - Planet 1

